

BEN MOHAMMED

Mohik

MP2i

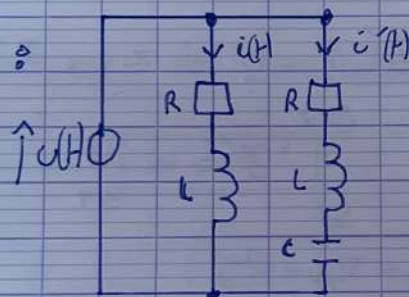
TD 0510

Exercice 6 (corrigé)

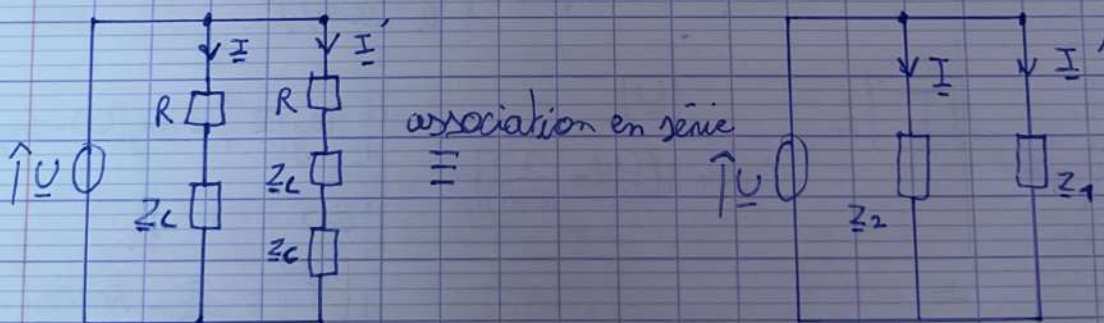
Grandeur temporelle	Nombre complexe	Amplitude complexe
$u(t) = U_m \cos(\omega t + \frac{\pi}{4})$	$\underline{u}(t) = \underline{U} e^{j\omega t}$	$\underline{U} = U_m e^{j\frac{\pi}{4}}$
$i(t) = I_m \cos(\omega t + \varphi)$	$\underline{i}(t) = \underline{I} e^{j\omega t}$	$\underline{I} = I_m e^{j\varphi}$
$i'(t) = I'_m \cos(\omega t + \varphi')$	$\underline{i}'(t) = \underline{I}' e^{j\omega t}$	$\underline{I}' = I'_m e^{j\varphi'}$

1)

On a :



son schéma complexe :



association en série :

$$\underline{Z}_1 = R + \underline{Z}_L + \underline{Z}_C$$

$$= R + jL\omega + \frac{1}{jC\omega}$$

$$\underline{Z}_2 = R + \underline{Z}_L$$

$$= R + jL\omega$$

Loi d'Ohm pour $\underline{I}$: $\underline{U} = \underline{Z}_2 \times \underline{I}$

$\Leftrightarrow$

$$\underline{I} = \frac{\underline{U}}{\underline{Z}_2}$$

on a alors, $\underline{I} = \frac{\underline{U}}{R + jL\omega}$

$\Rightarrow |\underline{I}| = \frac{|\underline{U}|}{|R + jL\omega|}$

$I_M = \frac{U_M}{\sqrt{R^2 + (L\omega)^2}}$

$\arg(\underline{I}) = \arg(\underline{U}) - \arg(R + jL\omega)$

$\Leftrightarrow$

$\varphi = \frac{\pi}{4} - \text{Arctan}\left(\frac{L\omega}{R}\right)$

Loi d'Ohm pour $\underline{I}'$: $\underline{U} = \underline{Z}_1 \times \underline{I}'$

$\Leftrightarrow$

$$\underline{I}' = \frac{\underline{U}}{\underline{Z}_1}$$

on a alors, $\underline{I}' = \frac{\underline{U}}{R + jL\omega + \frac{1}{jC\omega}}$

$$= \frac{\underline{U}}{R \left(1 + j \left(\frac{L\omega}{R} - \frac{1}{RC\omega}\right)\right)}$$

$|\underline{I}'| = \frac{|\underline{U}|}{\left|R \left(1 + j \left(\frac{L\omega}{R} - \frac{1}{RC\omega}\right)\right)\right|}$

$\arg(\underline{I}') = \arg(\underline{U}) - \arg\left(R + jL\omega + \frac{1}{jC\omega}\right)$

$\Leftrightarrow$

$\varphi' = \frac{\pi}{4} - \text{Arctan}\left(\frac{L\omega}{R} - \frac{1}{RC\omega}\right)$

$\Rightarrow$

$I_M' = \frac{U_M}{R \sqrt{1 + \left(\frac{L\omega}{R} - \frac{1}{RC\omega}\right)^2}}$

C'est l'opposé...

$$2) \text{ on a : } \psi = \varphi_0 - \varphi \quad \text{et} \quad \psi' = \varphi_0 - \varphi'$$

$$\psi = \varphi - \varphi_u \quad \psi' = \varphi' - \varphi_u$$

$$\psi = \varphi_0 - \varphi$$

$$\Leftrightarrow \psi = \arg(\underline{U}) - \arg(\underline{I})$$

$$\Leftrightarrow \psi = \arg\left(\frac{\underline{U}}{\underline{I}}\right)$$

$$\Leftrightarrow \psi = \arg(\underline{Z})$$

$$\Leftrightarrow \boxed{\psi = \text{Arctan}\left(\frac{L\omega}{R}\right)}$$

$$\psi' = \varphi_0 - \varphi'$$

$$\Leftrightarrow \psi' = \arg(\underline{U}) - \arg(\underline{I}')$$

$$\Leftrightarrow \psi' = \arg\left(\frac{\underline{U}}{\underline{I}'}\right)$$

$$\Leftrightarrow \psi' = \arg(\underline{Z}') = \arg(\underline{Z})$$

$$\Leftrightarrow \boxed{\psi' = \text{Arctan}\left(\frac{L\omega}{R} - \frac{1}{R\omega}\right)}$$

et on cherche les conditions sur L, C et ω pour avoir :

$$\boxed{\psi = -\psi'}$$

$\Leftrightarrow$

$$\text{Arctan}\left(\frac{L\omega}{R}\right) = -\text{Arctan}\left(\frac{L\omega}{R} - \frac{1}{R\omega}\right)$$

$\Leftrightarrow$

$$\text{Arctan}\left(\frac{L\omega}{R}\right) = \text{Arctan}\left(\frac{1}{R\omega} - \frac{L\omega}{R}\right)$$

$\Leftrightarrow$

$$\frac{L\omega}{R} = \frac{1}{R\omega} - \frac{L\omega}{R}$$

$\Leftrightarrow$

$$\frac{2L\omega}{R} = \frac{1}{R\omega}$$

$\times R\omega$ $\downarrow$

$$\Leftrightarrow \boxed{2LC\omega^2 = 1}$$